Under Actuator- DC motor Interfacing with 8051

Small DC Motor

DC motor Interfacing with 8051

- DC motor converts electrical energy in the form of Direct Current into mechanical energy.
- In the case of the motor, the mechanical energy produced is in the form of a rotational movement of the motor shaft.
- The direction of rotation of the shaft of the motor can be reversed by reversing the direction of Direct Current through the motor.
- The motor can be rotated at a certain speed by applying a fixed voltage to it. If the voltage varies, the speed of the motor varies.
- Thus, the DC motor speed can be controlled by applying varying DC voltage; whereas the direction of rotation of the motor can be changed by reversing the direction of current through it.
- For applying varying voltage, we can make use of the PWM technique.

- It translates electrical pulses to mechanical movement.
- It has only + and – terminal which rotates the motor in one direction.
- By reversing it motor change its direction.
- Maximum speed is indicated in rpm.
- Nominal Voltage will very from 1 to 150v. As we increase the dc voltage RPM will also be increased.

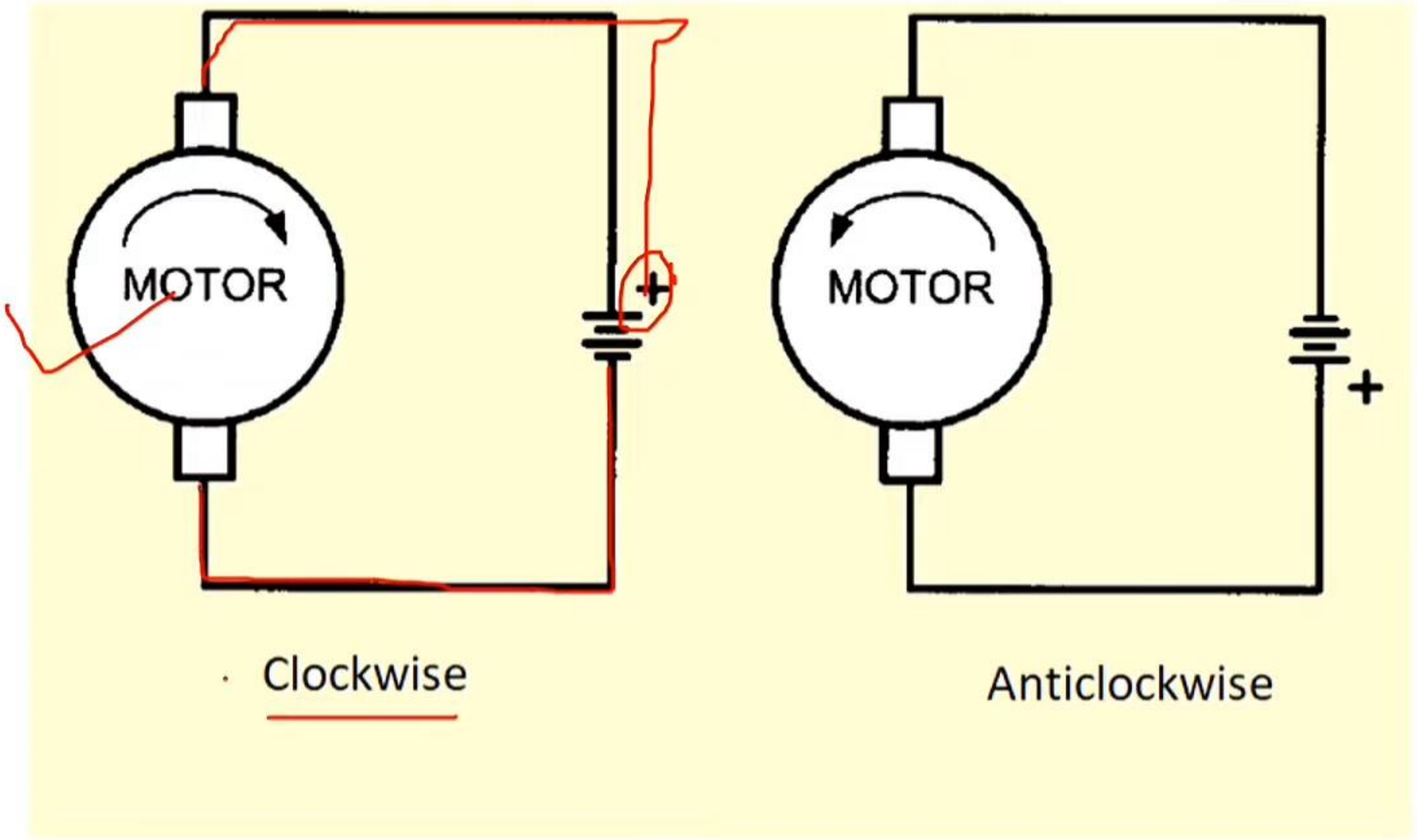
- So, as the load increases the RPM is decreased unless voltage or current is increased

- Which in turn increases the torque.

- With a fixed voltage as the load increases current(power) consumption will also be improved.

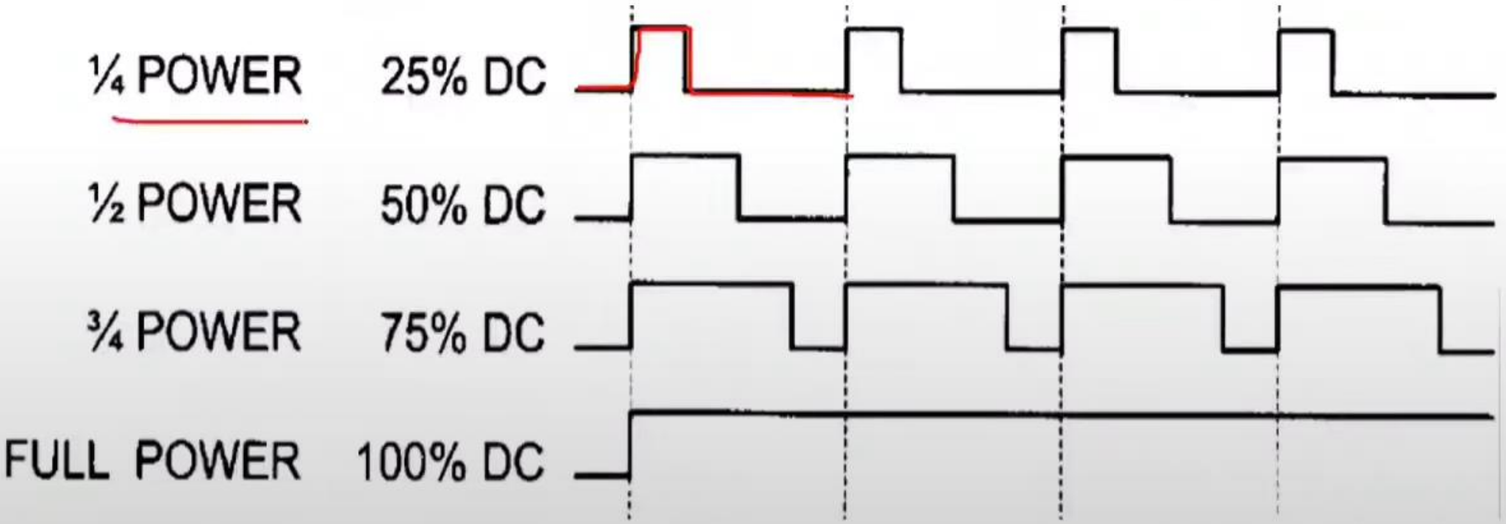
- If we overload the motor, it will halt and can damage the motor due to the heat.

Uni direction dc motor



Controlling the speed of the DC motor

- We can increase or decrease the motor speed by changing the amount of power to it by using PWM technique.



Calculate TH0 and TL0for 25% duty cycle for 20ms and fosc=11.0592 MHz

Case 1: Count directly given in the question

Case 2: If Frequency of oscillator is given, to calculate timer frequency

Timer frequency = Frequency of oscillator /12 OR

Count or Time – 12/Frequency of oscillator in secs

Count wrt timer frequency= Desired frequency/ Timer frequency

= in hex or decimal

Count = FFFF-count +1 -----hex

The formula used to find out the values of TH0 and TL0 is:

Ton = 25% of 20ms = 5ms

Toff = 75% of 20ms = 15ms

TH0 and TL0 value for Ton = 5ms

65,536 – (Time delay required/1.085μs) = 65,536 – (5ms/1.085μs)

= 0xEE00

So, TH0 = 0xEE and TL0 = 0x00 for 5ms ON time

65,536-(15ms/1.085μs) = 0xC9FF

So, TH0 = 0xC9 and TL0 = 0xFF for 15ms OFF time

For 50% TH0 = 0xDB and TL0 = 0xFF for 10ms ON time

TH0 = 0xDB and TL0 = 0xFF for 10ms OFF time

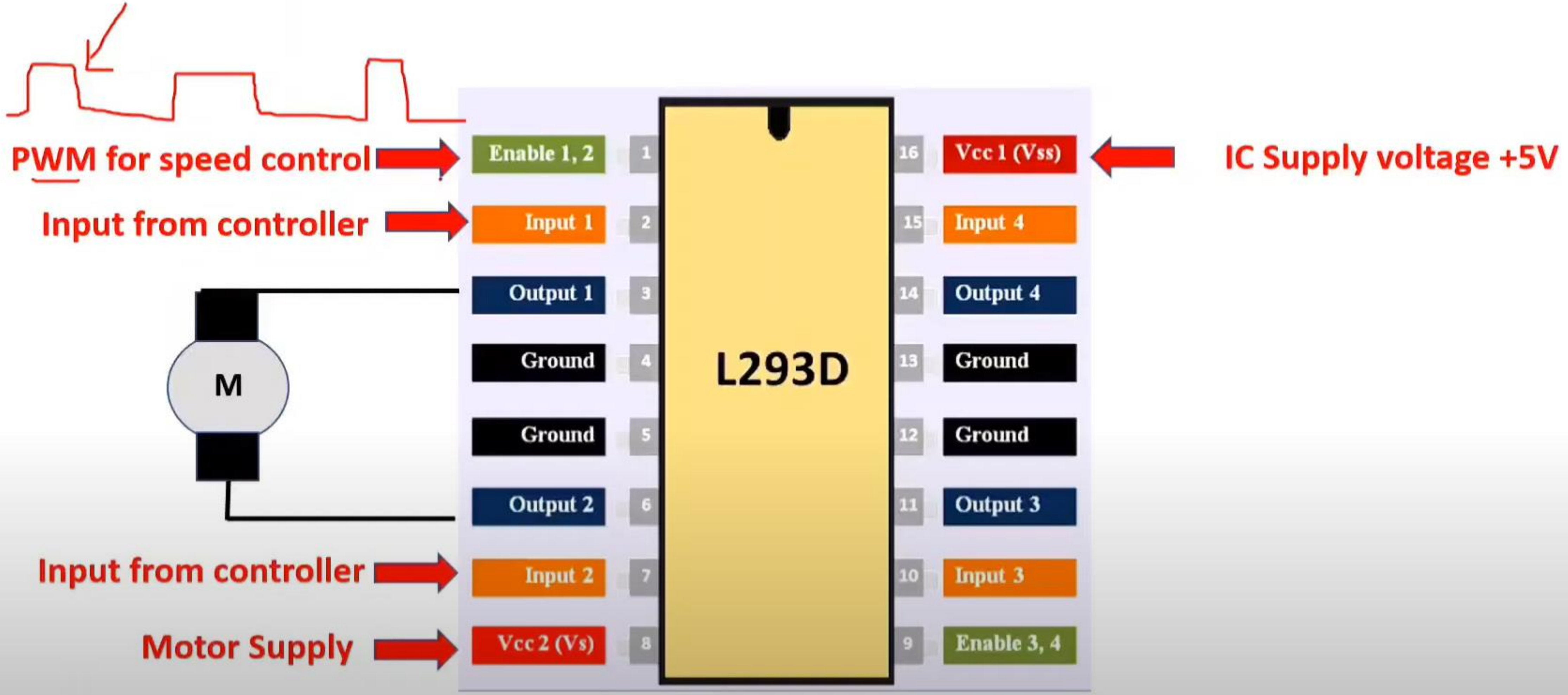
For 75% TH0 = 0xC9 and TL0 = 0xFF for 15ms ON time

TH0 = 0xEE and TL0 = 0x00 for 5ms OFF time

For 100% TH0 = 0xB7 and TL0 = 0xFF for 20ms ON time

TH0 = 0xFF and TL0 = 0xFF for 0ms OFF time

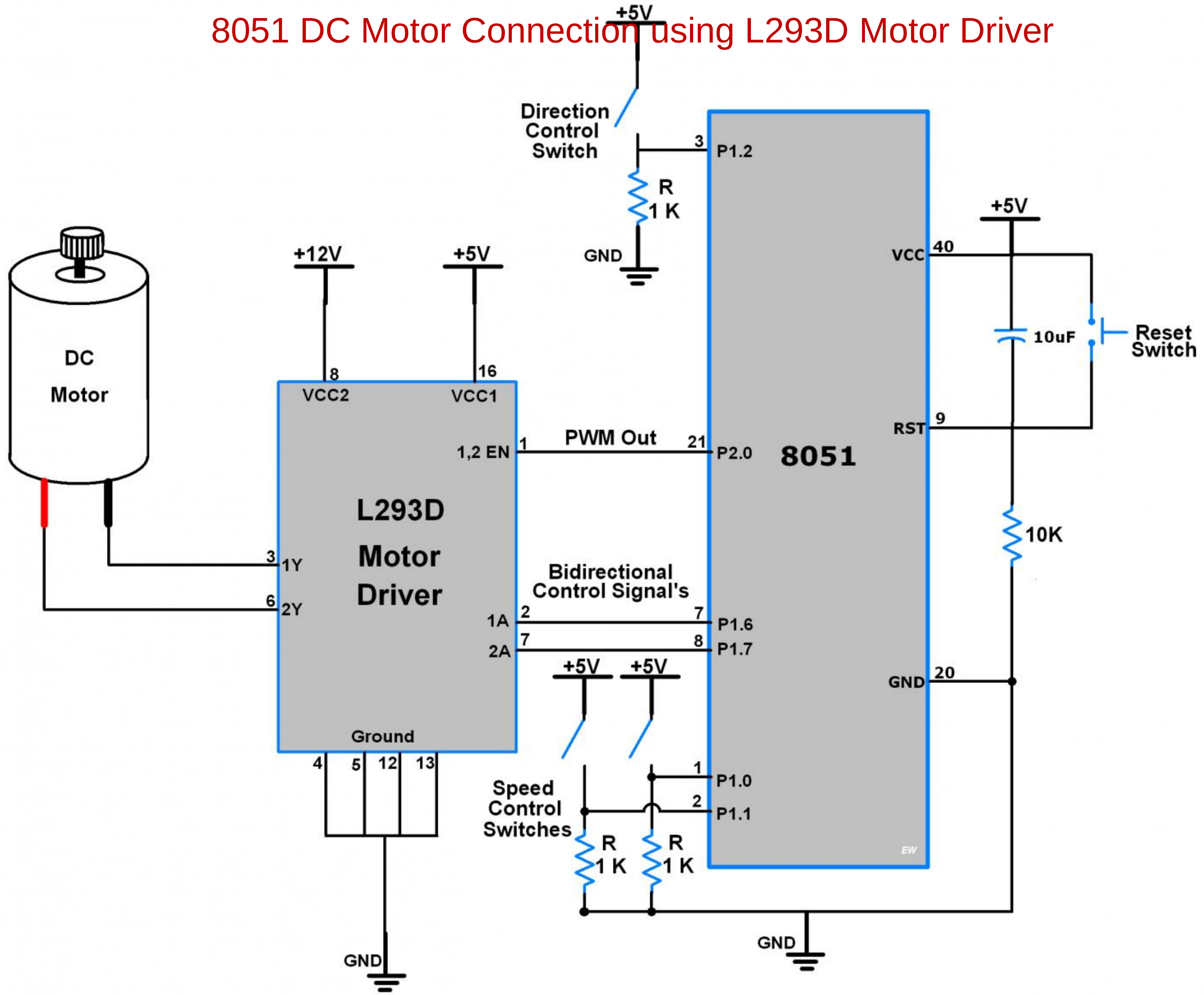
L293D DC motor driver IC

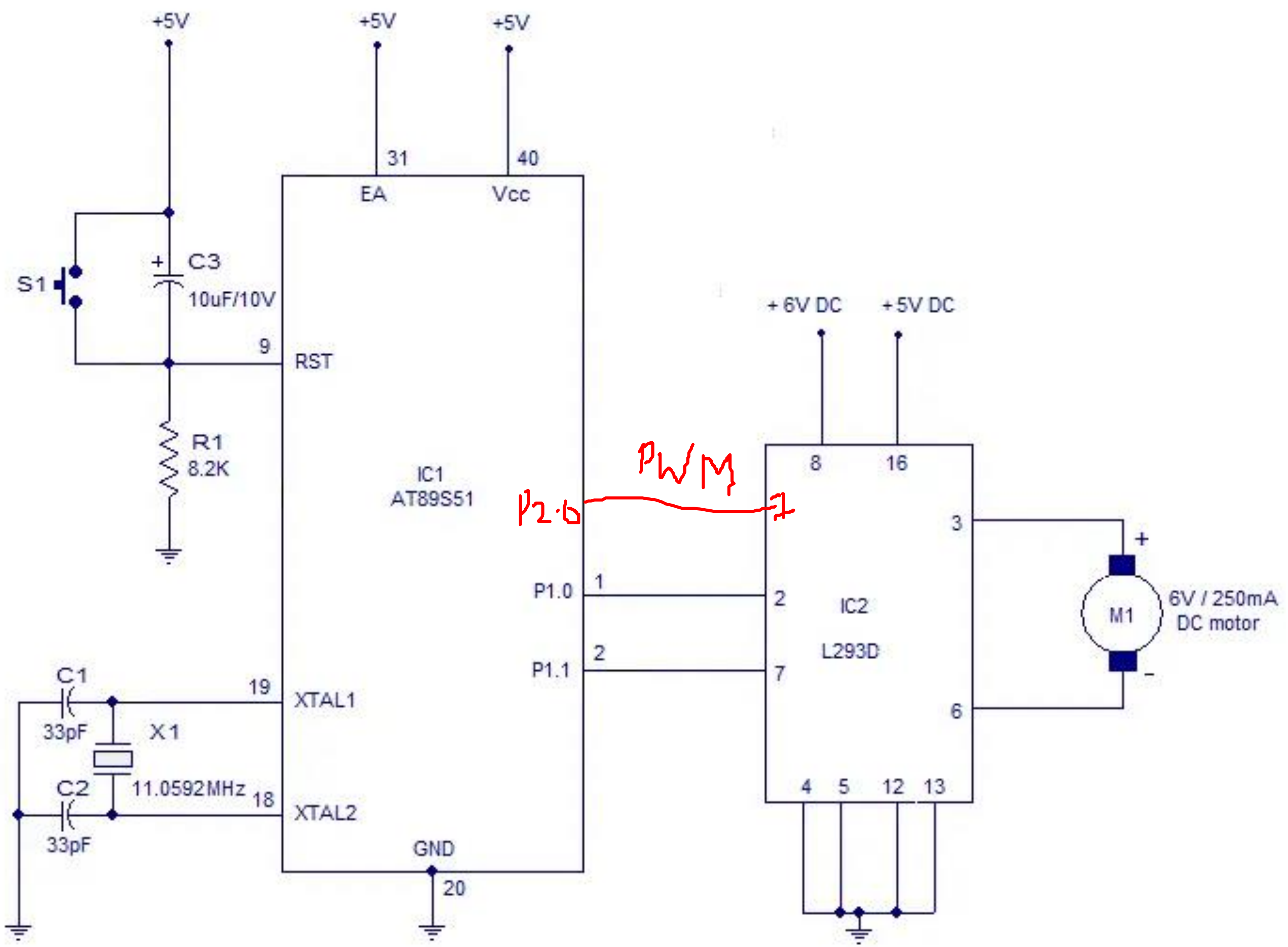


DC motor with 8051

- Let’s interface the DC motor with the AT89S52 microcontroller and control the DC motor speed by using the Speed Increment Switch and Speed Decrement Switch connected to the Microcontroller port and direction by using Direction Switch.
- We are going to use the L293D motor driver IC to control the DC motor movement in both directions. It has an in-built H-bridge motor drive.
- As shown in the above figure we have connected two toggle switches on the P1.0 and P1.1 pin of the AT89S52 microcontroller to change the speed of the DC motor by 10%.
- One toggle switch at pin P1.2 controls the motor's rotating direction.
- P1.6 and P1.7 pins are used as output direction control pins. It provides control to motor1 input pins of the L293D motor driver which rotate the motor clockwise and anticlockwise by changing their terminal polarity.
- And Speed of the DC Motor is varied through PWM Out pin P2.0.
- Here we are using the timer of AT89S52 to generate PWM.

8051 DC Motor Connection using L293D Motor Driver




+5V
+5V
+5V
31
40
EA
Vcc
S1
C3
10uF/10V
9
RST
R1
8.2K
IC1
AT89S51
P2.6
PWM
1
+ 6V DC
+ 5V DC
8
16
3
P1.0
1
2
IC2
L293D
M1
6V / 250mA
DC motor
P1.1
2
7
6
C1
33pF
X1
19
XTAL1
C2
11.0592MHz
18
XTAL2
33pF
GND
20
4
5
12
13

DC motor Interfacing with 8051

- ORG 00H // initial starting address
  MAIN: MOV P1,#00000001B // motor runs clockwise
- ACALL DELAY // calls the 1S DELAY
  MOV P1,#00000010B // motor runs anti clockwise
-
  ACALL DELAY // calls the 1S DELAY
  SJMP MAIN // jumps to label MAIN for repeating the cycle
-
  DELAY: MOV R4,#0FH
  WAIT1: MOV R3,#00H
  WAIT2: MOV R2,#00H
  WAIT3: DJNZ R2,WAIT3
  DJNZ R3,WAIT2
  DJNZ R4,WAIT1
  RET
  END
- https://www.circuitstoday.com/interfacing-dc-motor-to-8051